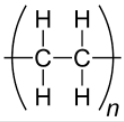
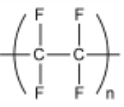
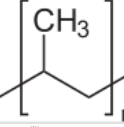
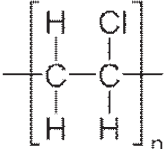
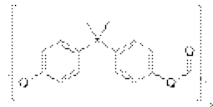


CHE 330 HW 1 Questions

Assigned 8/27/2025, Due 9/02/2025 11:59 PM

Polymers Basics

1. Read Chapter 1 of the course textbook ([Paul C. Hiemenz, Timothy P. Lodge - Polymer Chemistry-CRC Press \(2007\).pdf](#)) and the reading/viewing assignments that were posted on the UIC blackboard.
2. List five polymers that you can find in your daily life.
 - a. For each polymer, please provide the name of the polymer, the chemical structure of the monomer, and the chemical structure of the polymer using the right notation.
 - b. Explain why it is used for that application.

Polymer	Image	Use	Reason
Polythene		Packaging	flexible, durable. Moisture resistant
polytetrafluoroethylene (teflon)		Nonstick	chemical resistant, heat resistant
Polypropene		Bottles	lightweight, moisture resistant, durable
Polyvinyl chloride (PVC)		Pipes, insulation	durability, lightweight, resistant to corrosion
Polycarbonate		headlights	transparent, high impact resistance

3. Explain the difference between monomer, dimer, oligomer, polymer.

Monomer is a single unit molecule that forms larger macromolecules. Dimer is two chemically linked monomers. An oligomer is a molecule made of a few repeating units. Polymer is a macromolecule made of many repeating units.

Polymer Classification

4. Would you consider a polymer of MW=25,000 a low polymer or a high polymer? Why?

High polymer because the range for a low polymer is 1k-20k Da

5. Identify five polymers that can be found in nature

Starch, Cellulose, Proteins, Polysaccharides, DNA

6. Natural polymers are made by living organisms, while synthetic polymers are produced in the lab by chemical reactions. Explain a semi-synthetic polymer.

Semi-synthetic polymers are natural polymers that have been chemically modified.

7. List the 2 types of thermal responses of a polymer that we discussed in class. Describe them in your own words and list an example from class for each.

Thermoplastics when heated soften/melt, and when cooled they harden. They can be reshaped repeatedly with temperature.

Thermosets when heated irreversibly harden.

8. Which has a slower conversion, chain growth or step growth formation? (Hint: Use the graph)

Chain growth is faster

9. Briefly discuss 2 differences between a copolymer and a homopolymer.

A homopolymer is made up of a single type of monomer making it have a simple structure while copolymers are made up of two or more types of monomers. As a result, copolymers are more complex in structure.

10. Why do cellulose and starch, both made from glucose, have different properties? Describe how the differences in their structure at different levels (basic building blocks, chain arrangement, and overall 3D shape) affect properties like how well they dissolve in water, their crystallinity, and their rigidity.

Starch and cellulose are both made of glucose, but they differ in the type of bonds linking the glucose units. Starch contains alpha bonds which give it a coiled or branched structure. Cellulose contains beta bonds, which create straight, unbranched chains that can align closely and form hydrogen bonds. This results in a rigid, strong structure. So, starch is flexible and branched, while cellulose is strong and straight.